

Welcome to CSCI 004

This is an introductory to the subject of Computer Science in the form of a programming class. We will work through the basic structure and theory of modern programming concepts. Including (but not limited too) variables, language unique constructs, Types, and the mindset with which one should approach these concepts.

Objectives

1. Describe the software development life-cycle.
2. Describe the principles of structured programming.
3. Describe, design, implement, and test structured programs using currently accepted methodology.
4. Explain what an algorithm is and its importance in computer programming.

Teacher Info

My name is Evan Drake, I am a current researcher of computer science.

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Office Hours: M/W 10-11am, always at the butte college campus coffee shop.

Course outline:

Programming design tools

- what is Programming
 - environments
- pseudoCode

Introduction to Python

- Variables
- general intro to Logic for programming

Data Types

- primitive types are: Numbers, Characters, Boolean, Array, Tuples
 - Numbers can be expanded
- Most primitive of which is an array type
- RAM, the heap. Explain why types and variables are designed the way they are for machines / then maybe in language.

Functions

- function typing
- Why would we use a function

Coding conventions

- conventions include: style guide
- code comments: inline / block
- external docs like a wiki, or manual paper

Control flow

- if then else
- simple iterative loops

Extra material

Programming paradigms

- Procedural: systems of purely sequential instructions
- OOP: systems designed around objects, large structures describing traits rather than behavior

File IO

- **instream** / **outstream**, using **fstream** package
- sequential access

Passing parameters by value & by reference

- memory explanation of copy or reference value

Error handling

Principles of testing & designing test data

- unit tests
- integration tests
- divide and conquer strategies

Design principles

Class work

- zero late work policy
- we will have homework every week worth varying points.
- we may have quizzes these will never be graded for points.
- there will be extra credit available, we will talk about it on day one.

Butte College resources / Rules

- Academic guidelines
- Academic integrity
- Academic accommodations

Academic resources

- Center for academic success
- library website

Student technical resources

- Office 365 software
- Technical quickstart for students
- Canvas student guidelines